

Direction Sense

COMPLETE LEARNING GUIDE

What You'll Learn

Direction sense questions trace a person or object moving through a sequence of straight-line movements and turns, then ask for the final direction from the start, the shortest distance back to the start, or the direction of one point relative to another. The entire topic reduces to one skill: plotting each movement on an imaginary (or actually drawn) compass grid, with North up, East right, South down, West left, and correctly applying what 'left turn' and 'right turn' mean relative to the direction currently being faced — not relative to the compass. Most errors come from skipping the diagram and trying to track direction changes mentally, which falls apart after 3-4 turns. This sub-module covers compass basics and turn rules, net-displacement and the Pythagorean shortest-distance shortcut, shadow-based direction problems, clock-hand direction problems, multi-turn path tracing, and how to read final relative-direction questions (e.g., 'in which direction is the person from the starting point') off a completed diagram.

1. Compass Basics and Turn Rules

Fix North as up, South as down, East as right, West as left on your diagram. A RIGHT turn rotates your facing direction 90° clockwise from whatever you currently face; a LEFT turn rotates it 90° anticlockwise. Crucially, 'left' and 'right' are always relative to the direction the person is CURRENTLY facing, not fixed compass directions — someone facing South who turns 'left' ends up facing East, not West.

- A person starts facing North, turns right (now facing East), then turns right again (now facing South). Two successive right turns from North always land you facing South — a useful shortcut: two rights = a 180° reversal, four rights (or four lefts) return you to the original facing.

2. Plotting Movements and Net Displacement

Treat each straight-line movement as a vector in the current facing direction and plot it step by step on a simple x-y grid (East/West = x-axis, North/South = y-axis), keeping a running (x, y) coordinate from the start. The final answer to 'how far and in which direction from the start' is read directly off this final coordinate.

- A man walks 3 km North, then 4 km East. Plot: start (0,0) !' after North leg (0,3) !' after East leg (4,3). He is now 4 km East and 3 km North of the start. Straight-line distance from start = " $(3^2+4^2) = "$ 25 = 5 km (Pythagorean shortcut, since the two legs are perpendicular).

3. The Pythagorean Shortest-Distance Shortcut

Whenever the net East-West displacement and net North-South displacement are both non-zero, the straight-line ('shortest distance' or 'as the crow flies') distance from start to end is the hypotenuse of a right triangle formed by those two net displacements — always collapse a multi-leg path down to just its net East-West and net North-South totals before applying this, rather than trying to compute distance leg by leg.

- A man walks 6 km East, 8 km South, 6 km West. Net East-West: 6 East - 6 West = 0. Net North-South: 8 South. So he ends up exactly 8 km South of the start — the two equal East and West legs cancel out, and the shortest distance is simply 8 km (no Pythagorean step needed since one net component is zero).

4. Shadow-Based Direction Problems

These rely on one fixed fact: in the morning, the sun rises in the East, so a person's shadow falls toward the West (behind them, opposite the sun); in the evening, the sun is in the West, so the shadow falls toward the East. Establish the time of day from the question stem first, since it fully determines the shadow's direction.

- 'In the morning, Raj's shadow falls exactly behind Suresh.' Morning shadows point West, so Suresh is standing to the West of Raj (the shadow falls in the direction away from the sun, i.e., West, landing on/behind Suresh) — meaning Suresh is West of Raj.

5. Clock-Hand Direction Problems

Some direction questions use a clock face as a compass proxy: at 12 the minute/hour hand points 'North' (or another stated direction), and the question asks which direction the hour hand points at a different time, using the fact that a clock face is divided into 12 equal 30° sectors.

- At 12 o'clock, the hour hand points North. What direction does it point at 6 o'clock? The hour hand has moved exactly halfway around the 12-hour face (6 hours = 180°), so it now points South (directly opposite North).

6. Multi-Turn Path Tracing

For paths with 4 or more turns, the diagram is not optional — draw each leg with its length labeled, and mark the facing direction after every turn using the two-rights/two-lefts = reversal shortcut to catch arithmetic slips as you go, since a single missed turn direction compounds into a completely wrong final position.

- Starting facing East: turn right (facing South), walk 5 km; turn right (facing West), walk 5 km; turn right (facing North), walk 5 km. Three consecutive right turns from East land you facing North, and since each leg is equal length, this traces three sides of a square — the person now stands directly North of a point 5 km West of the start, i.e., diagonally related to the original position along the fourth, unwalked side.

7. Reading Final Relative-Direction Questions

Once the final (x, y) coordinate relative to the start is known, reading off the compass direction is a direct lookup: positive x = East, negative x = West, positive y = North, negative y = South, and a point with both a non-zero x and y component lies in an intercardinal direction (e.g., North-East, South-West) — state it as such rather than only giving distance.

Common Pitfalls

- Treating 'left' and 'right' turns as fixed compass directions instead of relative to the direction currently being faced.
- Skipping the diagram for a 3+ turn path and trying to track the final facing direction mentally.
- Applying the Pythagorean shortcut directly to multi-leg distances without first collapsing the path to net East-West and net North-South displacement.
- Getting the shadow direction backward — forgetting that a morning shadow points West (away from the Eastern sunrise), not toward it.
- Forgetting that two successive turns in the same rotational direction (left-left or right-right) always produce a 180° reversal, and re-deriving it the long way each time.
- Confusing 'distance traveled' (sum of all leg lengths) with 'shortest distance from start' (the straight-line hypotenuse), which the question usually asks for.
- Losing track of sign (positive/negative) on the x-y grid when a later leg reverses an earlier one, undercounting net displacement.

Key Takeaways

- Always plot movements on an (x, y) grid: East/West on the x-axis, North/South on the y-axis, tracking net displacement.
- Left and right turns rotate the CURRENT facing direction 90°; they are never fixed compass directions.
- Two consecutive same-direction turns (left-left or right-right) always produce a 180° reversal — a fast shortcut for multi-turn paths.
- Shortest distance from start = $\sqrt{(\text{net East-West})^2 + (\text{net North-South})^2}$ — only after collapsing the full path to net components.
- Morning shadows point West; evening shadows point East — this single fact resolves nearly all shadow-based direction questions.
- 'Total distance walked' and 'shortest distance from start' are different quantities — check which one the question is actually asking for.

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